

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil Engineering) SEMESTER - 6 Summer 2025 (Regular)

Course :Bachelor of Technology (Civil Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 6

Subject Code & Name: BTCVOE605K - DEVELOPMENT ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
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Q1. Objective type questions. (Compulsory Question)

12

- 1 Which of the following is NOT a core focus area in development engineering?
 - A. Socioeconomic development
 - B. Urban and rural development
 - C. Aerospace engineering
 - D. Scientific social research
- 2 Which of the following is a green building certification system in India?
 - A. BREEAM
 - B. GRIHA
 - C. ISO 9001
 - D. ASTM
- 3 What is the purpose of zoning in town planning?
 - A. To increase housing prices
 - B. To control land use and building density
 - C. To promote private businesses only
 - D. To eliminate public transport
- 4 What is the primary use of GIS in planning?
 - A. Entertainment
 - B. Web design
 - C. Mapping and layout preparation
 - D. Cooking
- 5 Which of the following best defines a 'research problem' in development engineering?
 - A. Any random issue in a community
 - B. A well-defined issue identified through fieldwork needing a development-based solution
 - C. An academic theory without practical implications
 - D. A financial audit of urban corporations

- 6 LEED certification is associated with which of the following?
 - A. Legal infrastructure
 - B. Road quality
 - C. Green building design and sustainability
 - D. Urban crime control
- 7 Which organization is responsible for energy standards often applied in sustainable building design?
 - A. ASHRAE
 - B. WHO
 - C. NASA
 - D. IMF
- 8 Which of the following is NOT typically part of a smart city infrastructure?
 - A. e-Governance
 - B. Urban mobility
 - C. Space missions
 - D. Waste management
- 9 Which element is considered a key performance benchmark in smart cities globally?
 - A. Population density
 - B. Amount of vehicles
 - C. ICT-based citizen services
 - D. Size of shopping malls
- 10 Which of these is a major challenge in rural public transport development?
 - A. Too many private airlines
 - B. Lack of demand
 - C. Sparse population and poor connectivity
 - D. High-speed train congestion
- 11 People's participation in rural development is essential because:
 - A. It reduces government funding
 - B. It ensures that local needs are addressed effectively
 - C. It delays project completion
 - D. It limits innovation
- 12 Which of the following is an interdisciplinary application of Geoinformatics?
 - A. Weather prediction
 - B. Land-use mapping
 - C. Disaster management
 - D. All of the above

Q2. Solve the following.

- A) Explain necessity and issues related to Development Engineering. 6
- B) What is socio- economic development? Explain the factors effecting development. 6

Q3. Solve the following.

- A) Explain any one standard for sustainable communities and certification for green building? 6
- B) Explain the formulation of research problems and explain various methods of data collection? 6

Q4. Solve Any Two of the following.

- A) Explain the objectives of smart city planning and also explain various Smart Solution? 6
- B) What is geoinformatics and explain the benefits and limitations of Geoinformatics? 6
- C) Explain master plan and density control? 6
- Q5. Solve Any Two of the following.
- A) Explain District Planning and its functions. 6
- B) What are the infrastructure elements of smart city planning? Explain dimensions of smart cities. 6
- C) Illustrate the Problems faced by the villagers pertaining to various sectors or existing schemes. 6
- Q6. Solve Any Two of the following.
- A) Explain the concept of sustainable community and its development? 6
- B) Explain different levels of town planning? 6
- C) Explain the application of geoinformatics? 6

*** End ***